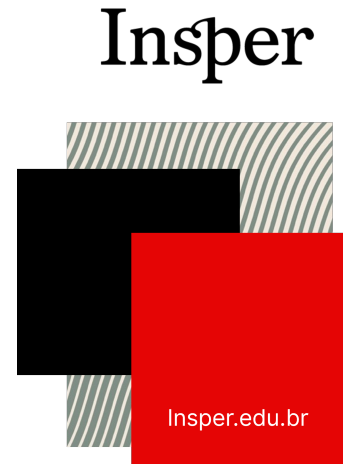
Economics of Organizations and Incentives

MPE – 4th term, 2026

Lecture 5: Repeated games



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Today

1. Finitely repeated games: unraveling, carrots and sticks.
2. The one-stage deviation principle.
3. Infinitely repeated games: payoffs, continuation strategies and subgame perfection.
4. Grim trigger and tit-for-tat.
5. Homework: tacit collusion in oligopoly.

Recap: repeated games

- A **repeated game** $G(T, \delta)$: the same stage game G is played in T periods; players observe past actions.
- Player i's payoff: $\sum_{t=1}^{T} \delta^{t-1} u_i^t$.
- A strategy says what to play in each period after each possible history; each period starts a new subgame.
- Solution concept: **subgame-perfect equilibrium**, which rules out non-credible threats.

Finitely repeated games

Finitely repeated prisoner's dilemma

Consider the prisoner's dilemma

		Player 2	
		m	f
Player 1	M	$4,\ 4$	$-1,\ 5$
	F	$5,\ -1$	$1,\ 1$

played repeatedly over T periods.

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Suppose that

- actions are publicly observable at the end of each period;
- the payoff in the repeated game is the sum of the payoffs received in all periods.

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Histories (subgames):

$$H_1 = \{\emptyset\}, \qquad H_t = \{Mm,\ Mf,\ Fm,\ Ff\}^{t-1}, \quad t = 2, \ldots, T.$$

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A strategy says what the player does at the start of the game ($h = \emptyset$) and after *every* possible history, including histories that will never happen in equilibrium.

The case $T = 2$: the last period

Let s be a subgame-perfect equilibrium. The subgame that follows an arbitrary history $h \in H_2 = \{Mm, Mf, Fm, Ff\}$ is:

	m	f
M	$u_1(h) + 4,\ u_2(h) + 4$	$u_1(h) - 1,\ u_2(h) + 5$
F	$u_1(h) + 5,\ u_2(h) - 1$	$u_1(h) + 1,\ u_2(h) + 1$

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which is equivalent to the stage game: adding the constant $u_i(h)$ does not change incentives.

Therefore $s(h) = (F, f)$ for every $h \in H_2$.

The case $T = 2$: the first period

Consider the incentives at the initial history, given that players choose $s(h)$ in the second period for every $h \in H_2$. Payoffs of player 1 (sum over the two periods):

	m	f
M	$4 + u_1(s(Mm))$	$-1 + u_1(s(Mf))$
F	$5 + u_1(s(Fm))$	$1 + u_1(s(Ff))$

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Since $s(h) = (F, f)$ for all $h \in H_2$, we add $u_1(F, f) = 1$ to every cell, which is equivalent to the stage game.

Thus $s_1(\emptyset) = F$. By symmetry, $s_2(\emptyset) = f$.

Unraveling

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- It also applies to any repeated game whose stage game has a **unique** Nash equilibrium.
- Bottom line: subgame perfection leads to an **unraveling** of incentives. The endgame effect propagates backwards and incentives for cooperation break down.
- Example: the "lame duck" president in American politics.

Stage-game equilibria and the repeated game

Proposition

If σ^* is a Nash equilibrium of the stage game G, then the strategy profile "play σ^* in every period, whatever the history" is a subgame-perfect equilibrium of $G(T, \delta)$.
If G has a **unique** Nash equilibrium, this is the **unique** subgame-perfect equilibrium of $G(T, \delta)$.

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Implication: to support non-Nash behavior (cooperation) in early periods of a finitely repeated game, the stage game needs **more than one** Nash equilibrium.

Stage games with multiple equilibria

Consider the following stage game, played in $T = 2$ periods:

		Player 2		
		m	f	r
Player 1	M	$4,\ 4$	$-1,\ 5$	$0,\ 0$
	F	$5,\ -1$	$1,\ 1$	$0,\ 0$
	R	$0,\ 0$	$0,\ 0$	$3,\ 3$

The overall payoff is the sum of the payoffs in the two periods.

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Stage-game Nash equilibria: (F,f), with payoffs $(1,1)$, and (R,r), with payoffs $(3,3)$.

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Stage games with multiple equilibria

Histories:

$$H_1 = \{\emptyset\}, \qquad H_2 = \{M, F, R\} \times \{m, f, r\}.$$

Strategies:

$$s_1 : H_1 \cup H_2 \to \{M, F, R\}, \qquad s_2 : H_1 \cup H_2 \to \{m, f, r\}.$$

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Consider the strategy profile s where, for each $h \in H_1 \cup H_2$,

$$s(h) = \begin{cases} (M, m) & \text{if } h = \emptyset, \\ (R, r) & \text{if } h = (M, m), \\ (F, f) & \text{if } h \neq (M, m),\ h \neq \emptyset. \end{cases}$$

Is s a subgame-perfect equilibrium?

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- **Last period:** for any history $h \in H_2$, $s(h)$ is a Nash equilibrium of the stage game, and the subgame that follows h is equivalent to the stage game.
- It remains to check incentives at the history $h = \emptyset$.
- Incentives of player 1 at $h = \emptyset$ (the best deviation is F):

$$\underbrace{u_1(M, m) + u_1(s(M, m))}_{4+3\,=\,7} \;\geq\; \underbrace{u_1(F, m) + u_1(s(F, m))}_{5+1\,=\,6}.$$

By symmetry, the same holds for player 2. So s is a subgame-perfect equilibrium.

The one-stage deviation principle

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In a finitely repeated game, a strategy profile s is a subgame-perfect equilibrium if and only if no player i can gain by deviating from $s_i(h)$ at a single history h and conforming to s_i afterwards.

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- Intuition: if deviating at several histories is profitable, undo the deviations one at a time, starting from the last one; at some point, a single deviation must already be profitable on its own.
- It greatly simplifies the analysis: instead of all possible deviations, we check only **one-stage deviations**, history by history.

With discounting

If payoffs are discounted, $u_i^1 + \delta\, u_i^2$, the incentive constraint at $h = \emptyset$ becomes

$$4 + 3\delta \;\geq\; 5 + \delta \iff \delta \geq \tfrac{1}{2}.$$

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- The good equilibrium (R, r) is the **carrot**; the bad equilibrium (F, f) is the **stick**.
- The punishment is **credible**: the stick is itself a Nash equilibrium of the subgame.
- The future must matter enough: the difference between carrot and stick $(3 - 1)$, weighted by δ, must outweigh the short-run gain from deviating $(5 - 4)$.

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- This requires multiple equilibria in the stage game. Otherwise, the endgame effect leads to unraveling, and incentives for cooperation break down.
- With a unique stage-game equilibrium, the known last period destroys cooperation. **What if there were no last period?**

Infinitely repeated games

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- The problem with the finitely repeated prisoner's dilemma comes from the **common knowledge** that the game has a fixed and finite number of periods.
- If the stage game has a unique Nash equilibrium, sequential rationality in the last period T leads to unraveling.
- What happens if we remove this problem by assuming that the game has **no final period**?

Infinitely repeated games

Stage game:

$$G = \big(I, (A_i), (u_i)\big),$$

where A_i is the set of actions of player i, $A = \times_{i \in I} A_i$, and $u_i : A \to \mathbb{R}$ is her stage-game payoff.

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Strategies:

$$s_i : \bigcup_{t \geq 1} H_t \to A_i, \qquad i \in I.$$

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Outcomes:

$$H_\infty = A \times A \times A \times \cdots \quad (\infty \text{ times})$$

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Each strategy profile s determines a unique outcome

$$\left(a^1(s), a^2(s), \ldots, a^t(s), \ldots\right) \in H_\infty$$

in the repeated game:

$$\begin{aligned} a^1(s) &= s(\emptyset), \\ a^2(s) &= s\left(a^1(s)\right), \\ &\vdots \\ a^t(s) &= s\left(a^1(s), \ldots, a^{t-1}(s)\right), \ \ldots \end{aligned}$$

Payoffs in infinitely repeated games

Payoffs:

$$U_i(s) := (1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}u_i\big(a^t(s)\big), \qquad i \in I,$$

where $\delta \in (0,1)$ is the discount factor and $\big(a^1(s), a^2(s), \dots\big)$ is the outcome of s.

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The normalization factor $(1-\delta)$ puts repeated-game payoffs on the same scale as stage-game payoffs: if a player gets k in every period, her repeated-game payoff is

$$(1-\delta)\big(k + \delta k + \delta^2 k + \delta^3 k + \cdots\big) = (1-\delta)\,\frac{k}{1-\delta} = k.$$

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We denote this repeated game by G^δ.

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- **Discounting future payoffs** is a natural economic assumption when payoffs occur over time.
- **Infinite play** is theoretically possible, but not realistic in most settings.
- **Practical interpretation:** discounting can reflect *uncertainty about the future*.
- In each period, the game continues to the next period with probability $\delta < 1$; with probability $1 - \delta$, the relationship ends for exogenous reasons.
- What matters is not that the game lasts forever, but that **no one knows which period is the last one**.

Subgames and continuation strategies

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Definition (continuation strategy)

Let s_i be a strategy of player i and h a history. The **continuation strategy** induced by s_i at h is

$$s_i|_h(h') := s_i(h, h'), \quad \text{for all } h' \in \bigcup_t H_t.$$

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Proposition

A strategy profile s is a subgame-perfect equilibrium of G^δ if and only if, for any history h, $s|_h$ is a Nash equilibrium of G^δ.

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Characterizing subgame-perfect equilibrium

Proposition

In an infinitely repeated game G^{δ}, a strategy profile $s = (s_1, s_2, \ldots, s_n)$ is a **subgame-perfect equilibrium (SPE)** if and only if there is no player i and no single history h_{t-1} from which player i would gain by deviating from $s_i(h_{t-1})$.

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Interpretation: to verify that a strategy profile is an SPE, it is enough to ensure that no player has a profitable **one-stage deviation** at any point in the game.

Using the one-stage deviation principle

Idea:

- The **one-stage deviation principle** states that *a strategy profile is an SPE if no player gains from deviating at any single history*.
- This greatly simplifies checking equilibria: instead of considering all possible future deviations, we check only **one-period deviations**.

Using the one-stage deviation principle

Idea:

- The **one-stage deviation principle** states that *a strategy profile is an SPE if no player gains from deviating at any single history*.
- This greatly simplifies checking equilibria: instead of considering all possible future deviations, we check only **one-period deviations**.

Formally: s is an SPE if and only if, for each history h, player i and action $a_i \in A_i$,

$$U_i(s|_h) \;\geq\; (1-\delta)\, u_i\big(a_i, s_{-i}(h)\big) + \delta\, U_i\big(s|_{(h,(a_i,s_{-i}(h)))}\big).$$

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Key implication: even though there are infinitely many histories, the equilibrium condition reduces to local, single-stage checks.

Application: the repeated prisoner's dilemma

In the repeated prisoner's dilemma:

- there are infinitely many possible histories;
- however, for many strategies they can be grouped into a few relevant **states**, for example:
  1. no deviation so far: players play (M, m);
  2. a deviation has occurred: players play (F, f).

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- there are infinitely many possible histories;
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  2. a deviation has occurred: players play (F, f).

We can describe such a strategy as an **automaton**: a set of states, an action in each state, and transitions between states that depend on what was played. To verify SPE, we only need to check one-stage deviations in each state.

Grim trigger and tit-for-tat

Prisoner's dilemma: grim trigger

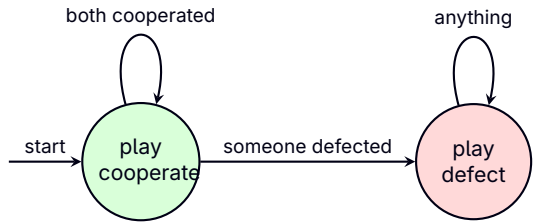
Start by cooperating. Cooperate **as long as everyone has cooperated**; if anyone ever defects, **defect forever**:

$$s(h) = \begin{cases} (M, m) & \text{if } h = \emptyset \text{ or } h = (Mm, Mm, \ldots, Mm), \\ (F, f) & \text{otherwise.} \end{cases}$$

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The "defect" state is absorbing: cooperation never resumes.

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Cooperative state ($h = \emptyset$ or $h = (Mm, \dots, Mm)$). The incentive constraint of player 1 is

$$U_1(s|_h) \;\geq\; (1-\delta)\, u_1(F, m) + \delta\, U_1\big(s|_{(h,Fm)}\big) \iff 4 \;\geq\; (1-\delta)\, 5 + \delta \cdot 1 \iff \delta \geq \tfrac{1}{4}.$$

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Punishment state (any other history). The incentive constraint is

$$1 \;\geq\; (1-\delta)(-1) + \delta \cdot 1,$$

which always holds: (F,f) is a Nash equilibrium of the stage game.

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Let us apply the one-stage deviation principle.

Cooperative state ($h = \emptyset$ or $h = (Mm, \dots, Mm)$). The incentive constraint of player 1 is

$$U_1(s|_h) \;\geq\; (1-\delta)\, u_1(F,m) + \delta\, U_1\big(s|_{(h,Fm)}\big) \;\iff\; 4 \;\geq\; (1-\delta)\, 5 + \delta \cdot 1 \;\iff\; \delta \geq \tfrac{1}{4}.$$

Punishment state (any other history). The incentive constraint is

$$1 \;\geq\; (1-\delta)(-1) + \delta \cdot 1,$$

which always holds: (F,f) is a Nash equilibrium of the stage game.

Thus, grim trigger is an SPE if and only if $\delta \geq \frac{1}{4}$.

Prisoner's dilemma: tit-for-tat

Tit-for-tat: start by cooperating and then **copy what the opponent did in the previous period**:

$$s_i(h) = \begin{cases} M & \text{if } h = \emptyset, \\ a_{-i} & \text{if } h = (\ldots, (a_1, a_2)), \end{cases}$$

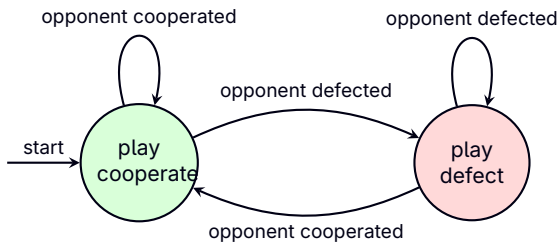
with m in place of M for player 2.

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Tit-for-tat: a single deviation echoes forever

Both players follow tit-for-tat, but player 1 defects *once*, in period 3:

Period	1	2	3	4	5	6	⋯
Player 1	M	M	F	M	F	M	⋯
Player 2	m	m	m	f	m	f	⋯

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Each player copies what the other did in the previous period, so a single deviation turns into an endless cycle of retaliation: $Fm,\ Mf,\ Fm,\ Mf, \ldots$

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Player 1's payoff along this cycle, starting with Fm:

$$(1-\delta)\left(5-\delta+5\delta^2-\delta^3+\cdots\right)=\frac{5-\delta}{1+\delta}.$$

Tit-for-tat: is it an SPE?

With tit-for-tat, what matters is only the last action profile. One-stage deviation checks for player 1:

Last profile	Follow tit-for-tat	Deviate once	Condition
$\emptyset$ or Mm	4	$\frac{5-\delta}{1+\delta}$	$\delta \geq \frac{1}{5}$
Mf	$\frac{5-\delta}{1+\delta}$	4	$\delta \leq \frac{1}{5}$

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No δ satisfies all four conditions: in this game, **tit-for-tat is not an SPE for any** δ.

- Patience helps with incentives **on** the path of s (first row).
- But patience hinders incentives **off** the path: punishing the opponent also punishes yourself.

Homework: tacit collusion

Homework: tacit collusion in oligopoly

Two firms play the Cournot game from Lecture 2 repeatedly, forever, with discount factor δ: inverse demand $P = 100 - q_1 - q_2$ and cost $C(q_i) = 10\,q_i$.

1. Find the Cournot-Nash outputs and profits of the stage game. Find the **cartel** outcome, in which each firm produces half of the monopoly output, and the profit of each firm.
2. Suppose the rival produces the cartel output. What is the best deviation, and the profit it yields?
3. Write the **grim-trigger** strategy: produce the cartel output as long as both firms have done so; otherwise, produce the Cournot-Nash output forever. Use the one-stage deviation principle to find the values of δ for which it is a subgame-perfect equilibrium.
4. Show that the threshold does not depend on the demand and cost parameters: redo item 3 with $P = a - b(q_1 + q_2)$ and marginal cost $c < a$.
5. Which market features make tacit collusion easier? Relate them to item 3.

Wrap-up

- With a unique stage-game equilibrium, a known last period makes cooperation unravel.
- With two stage-game equilibria, a carrot and a stick can sustain cooperation in early periods, if players are patient enough.
- One-stage deviation principle: to check subgame perfection, check single deviations history by history.
- Infinitely repeated games: with no known last period, grim trigger sustains cooperation in the prisoner's dilemma if $\delta \geq \frac{1}{4}$; tit-for-tat fails off the path.
- Homework: tacit collusion in a repeated Cournot duopoly.